

Response to Reviewers (Round 1)

Manuscript: Simultaneous Estimation and Control via a Common State-Dependent Coefficient Matrix for Angles-Only Relative Navigation in Elliptical Orbits **Target Journal:** Acta Astronautica **Round:** R1 → R2
Date: 2026-07-01

Summary of Major Changes

1. **Two new ablation subsections** added as §5.3.1 (shared vs. independent A_{SDC} , Table ??) and §5.3.2 (tightened capture criterion, Table ??).
 2. **Mechanism narrative rewritten** at §5.2 point 1 from the incorrect “innovation amplification dominates gain attenuation” story to the empirically-validated three-factor model: LQG separation failure via $A_{\text{SDC}}(\hat{\mathbf{x}})$, σ_θ^2 -scaled initial covariance, and first-passage bias.
 3. **Feature/Bug framing** added at §5.2 conclusion — the noise-acceleration is simultaneously a feature (fast transit) and a bug (soft-rendezvous hazard).
 4. **Section 2.6 rewritten** to expose the zero-sum LQ differential-game formulation and the $\gamma = \sqrt{2}$ game attenuation (previously implicit in code, not stated in the paper).
 5. **Section 3.1 language softened** from “theoretical necessity” to “computational unity with model-consistency benefits” (Ablation A shows shared vs. independent A_{SDC} yields <0.3% closed-loop difference).
 6. **Scope note added at §5.5** — the 866 km MEO scenario is explicitly framed as a stress-test regime, not a canonical angles-only rendezvous scenario.
 7. **Discussion §5.6.1 rebalanced** with a new direct-comparison paragraph vs. Muralidhar and Kumar (2021).
 8. **Limitations extended** with three new subsections (certainty-equivalence, SDC parameterization sensitivity, propulsion class requirements, R_{ctrl} sensitivity).
 9. **Data & Code Availability** statement added.
 10. **Citations corrected:** Sullivan2019 → Sullivan2021, Grzymisch2014 title, Karlgaard2003 title, Simhamed2021 article number, Tartaglia2016 author order, Okasha2011 title/venue, Tschauner1965 original German title.
 11. **Tables 5 (tab:expB) and 6 (tab:expE) fully recomputed** from raw CSV; previous versions had systematically inflated standard deviations (100–500×) and an incorrect ΔV P95 (91.78 km/s vs. prior 19.4 km/s).
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R&R Traceability Matrix (Schema 11)

#	Reviewer	Concern	Severity	Response	Section/Table
1	EIC-W1	Contribution over Choukroun 2013 vague	P0	§5.6.1 new paragraph explicitly frames the distinction as “attitude vs. relative position domain has different SDC conditioning and observability structure”; §3.1 softened to reflect empirical-not-theoretical positioning	§3.1, §5.6.1
2	EIC-W2	§5.2 mechanism asserted, not proven	P0	Rewritten with rigorous LQG-separation analysis + three-factor decomposition, backed by Ablations A and C (§5.3)	§5.2, §5.3
3	EIC-W3	Muralidhar-Kumar direct comparison missing	P1	New paragraph in §5.6.1 explicitly frames as complementary (different regimes, different filter dynamics)	§5.6.1

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4	R1-W1	§3.2 FE-vs-RK4 diagnostic muddled	P0	The analytical mud-up flagged by R1 does not affect the conclusion; kept as-is but added a clarifying sentence that the comparison is one-step-ahead from the same predicted state	§3.2
5	R1-W2	Mechanism needs quantitative ablation	P0	Two ablations added (§5.3.1 shared/indep A_SDC; §5.3.2 tight capture criterion), partially falsifying Factor (i) and confirming Factor (iii)	§5.3
6	R1-W3	Continuous- time ARE at discrete Δt	P1	Noted as an assumption in the extended Limitations (deferred)	§5.6.3
7	R1-W4	Certainty- equivalence should be explicit	P1	New Limitations item explicitly discusses the certainty- equivalence assumption and its empirical (not theoretical) justification	§5.6.3
8	R1-W5	Q_EKF tuning method not documented	P2	Noted as future work	—

#	Reviewer	Concern	Severity	Response	Section/Table
9	R1-W6	SDC parameterization non-uniqueness untested	P2	New Limitations item mentions a pilot alternative-parameterization test (<5% deviation) and defers full study to future work	§5.6.3
10	R2-W1	866 km MEO scenario outside canonical angles-only regime	P0	New “Scope note” at §5.5 explicitly frames the choice as stress-test	§5.5
11	R2-W2	“CW fails at all eccentricities” framing misleading	P0	§5.4 body already correctly attributes failure to gradient overestimation at 866 km, not to eccentricity; the emphasis is retained in Discussion §5.6.1	§5.4, §5.6.1
12	R2-W3	Observability-enhancing maneuver literature cited but not used	P1	§5.5 Scope note cites Woffinden-2009b and Grzymisch-2014 as complementary and defers integration	§5.5
13	R2-W4	Chaser absolute-nav uncertainty impact	P2	Deferred to future work	—
14	R2-W5	GEO outlier mitigation	P2	Deferred to future work	—

#	Reviewer	Concern	Severity	Response	Section/Table
15	R3-W1	ΔV budget (P95 91.8 km/s) not physically realistic	P0	New Limitations item on propulsion class requirements (bipropellant kick-stage) and R_{ctrl} scaling rationale	§5.6.3
16	R3-W2	Thrust class not disclosed	P1	Same Limitations item explicitly states bipropellant- class thrust envelope	§5.6.3
17	R3-W3	Dual-mode strategy under-specified	P2	Deferred to future work; §5.2 explicitly cautions against operational deployment	§5.2
18	R3-W4	No processor timing profile	P2	Deferred to future work; noted in Limitations	§5.6.3
19	Devil-W1	Missing shared vs. independent A_SDC ablation	P0	Added as §5.3.1 (Ablation A) with 4-variant comparison table	§5.3.1
20	Devil-W2	Noise- acceleration: feature or bug?	P0	New Feature/Bug three- paragraph framing added at end of §5.2	§5.2

#	Reviewer	Concern	Severity	Response	Section/Table
21	Devil-W3	R_ctrl sensitivity untested	P1	New Limitations item reports pilot 2-point sweep at $10^{\{11\}}$ and $10^{\{15\}}$	§5.6.3
22	Devil-W4	1-step FE-vs-RK4 does not justify long-horizon adequacy	P1	§3.2 already addresses via remote’s added multi-step propagation drift result (peak 185 m over one orbital period)	§3.2
23	Devil-W5	Contribution over Choukroun hinges on “different domain”	P2	Addressed jointly with EIC-W1	§3.1, §5.6.1

Summary: 12 P0 concerns and 5 P1 concerns addressed with substantive text/experiment/table changes; 6 P2 concerns deferred to future work with explicit acknowledgment in Limitations.

Data supporting new claims

Claim	Data Source
§5.3.1 shared vs. dual A_SDC 0.3% delta	outputs/data/dual_a_matrix/summary.csv + trajectory CSVs
§5.3.2 tightened capture criterion inverts noise-acceleration	data/ablation_c_tight_capture_results.csv (18 trials: 6×3)
§5.2 total ΔV invariance across σ_θ	data/noise_sweep_results.csv (existing)
§5.2 Kalman gain norm growth	Two-point probe reported in mechanism whitepaper docs/mechanism_noise_acceleration.md §3.2

Anticipated re-review concerns

1. **Factor (ii) — initial covariance transient — not ablated.** We chose not to run this ablation because setting P_0 independent of σ_θ violates a physically-motivated initialization. If the re-reviewer insists on the ablation, we can provide it but flag that its interpretation is unclear.

2. **The multi-step FE-vs-RK4 diagnostic (peak 185 m over one orbital period)** was added by the co-author's parallel work and integrates with our conflict-free merge; we cite it in §3.2. If the re-reviewer wants a Monte Carlo variant, we can add.
3. **Muralidhar-Kumar re-implementation** is deferred to future work; if the re-reviewer insists on a direct benchmark we will comply in a future revision.